Solution

(i) Since X denotes the number of success, X can take the values $0,1,2,...,10$

The probability for success is $p=\frac{1}{4}$ and for failure $q=1-p=\frac{3}{4}$, and $n=10$.

Therefore X follows the binomial distribution $X\sim B\left(10,\frac{1}{4}\right)$.

This gives, $$f(x)=\binom{10}{x}\left(\frac{1}{4}\right)^{x}\left(\frac{3}{4}\right)^{10-x},\qquad x=0,1,2,...,10$$

(ii) Probability for seven correct answers is

$$P(X=7)=f(7)=\binom{10}{7}\left(\frac{1}{4}\right)^{7}\left(\frac{3}{4}\right)^{10-7}=120\left(\frac{3^3}{4^{10}}\right)$$

Probability that the student will get seven correct answers is $120\left(\frac{3^3}{4^{10}}\right)$.

(iii) Probability for at least one correct answer is

$$P(X\geq 1)=1-P(X<1)=1-P(X=0)$$
$$=1-\binom{10}{0}\left(\frac{1}{4}\right)^{0}\left(\frac{3}{4}\right)^{10}=1-\left(\frac{3}{4}\right)^{10}.$$

Probability that the student will get for at least one correct answer is $1-\left(\frac{3}{4}\right)^{10}$.

Example 11.21

The mean and variance of a binomial variate X are respectively 2 and 1.5. Find

(i) $P(X=0)$ (ii) $P(X=1)$ (iii) $P(X\geq 1)$

Solution

To find the probabilities, the values of the parameters *n* and *p* must be known.

Given that

$$\text{Mean}=np=2 \text{ and variance } =npq=1.5$$

This gives $$\frac{npq}{np}=\frac{1.5}{2}=\frac{3}{4}$$

$$q=\frac{3}{4} \text{ and } p=1-q=1-\frac{3}{4}=\frac{1}{4}$$

$np=2$, gives $n=\frac{2}{p}=8$. Therefore $X\sim B\left(8,\frac{1}{4}\right)$.

Therefore the binomial distribution is

$$P(X=x)=f(x)=\binom{8}{x}\left(\frac{1}{4}\right)^{x}\left(\frac{3}{4}\right)^{8-x}\qquad x=0,1.2,...8$$

(i) $$P(X=0)=f(0)=\binom{8}{0}\left(\frac{1}{4}\right)^{0}\left(\frac{3}{4}\right)^{8-0}=\left(\frac{3}{4}\right)^{8}$$

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